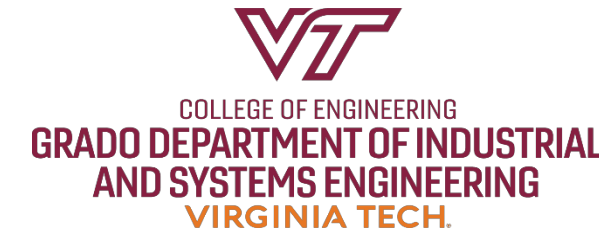


Effects of Agent Interactivity and Empathy Style on Drivers' Performance and Experience



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Introduction

- Drivers frequently experience frustration or anger in real-world traffic, which negatively affects driving performance and safety.
- Empathic in-vehicle agents have shown potential in helping regulate these emotions.
- In study I, we compared *cognitive vs. affective empathy* delivered by a voice-only agent:
 - Cognitive empathy: the ability to take the perspective of others and understand their thoughts, feelings, and mental state
 - Affective empathy: the ability to share and emotionally mirror the emotions of others
- In study II, we compared the impact of *embodied robots vs. voice-only agents* on driver behavior and emotion.
- Conversational LLM agents may alter how empathy is perceived. Study III compares their impact on driver behavior and emotion to that of purely scripted dialogue.

Research Objectives

To investigate whether empathic utterances generated by an LLM from an embodied in-vehicle agents in frustrating driving contexts influence drivers' performance and their affective states.

Research Questions:

- RQ1** How does *conversation* (pre-scripted vs. LLM-powered agents) influence *drivers' perceptions* of the in-vehicle agent (e.g., perceived empathy, authenticity, social presence, and trust)?
- RQ2** How do different empathy styles (cognitive, affective, and non-empathic) and their interaction with conversational interactivity shape drivers' emotional responses during frustrating driving events?
- RQ3** How do empathy styles and conversational interactivity affect *drivers' driving performance*, particularly in high-demand, safety-critical situations?

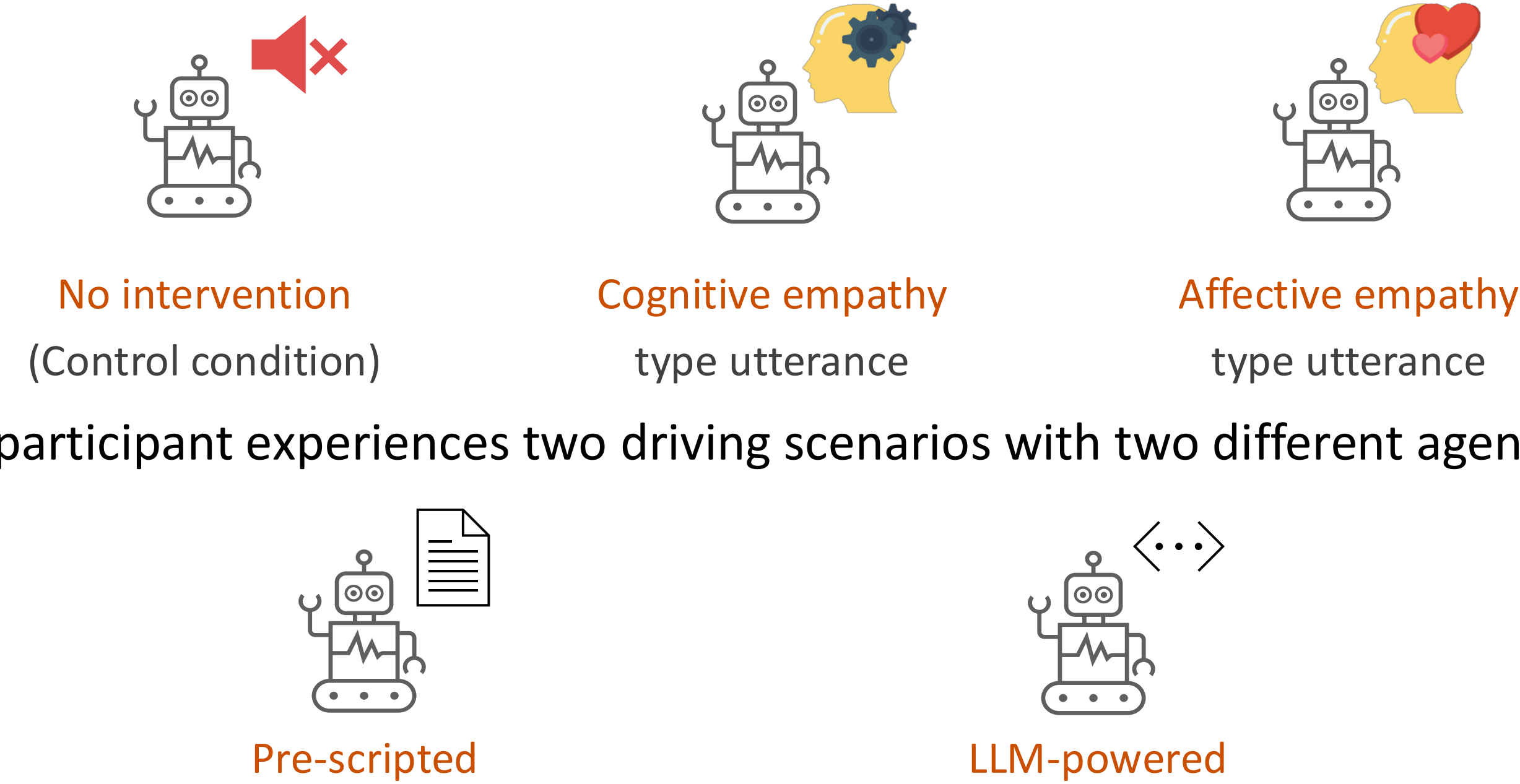
Methods

Participants

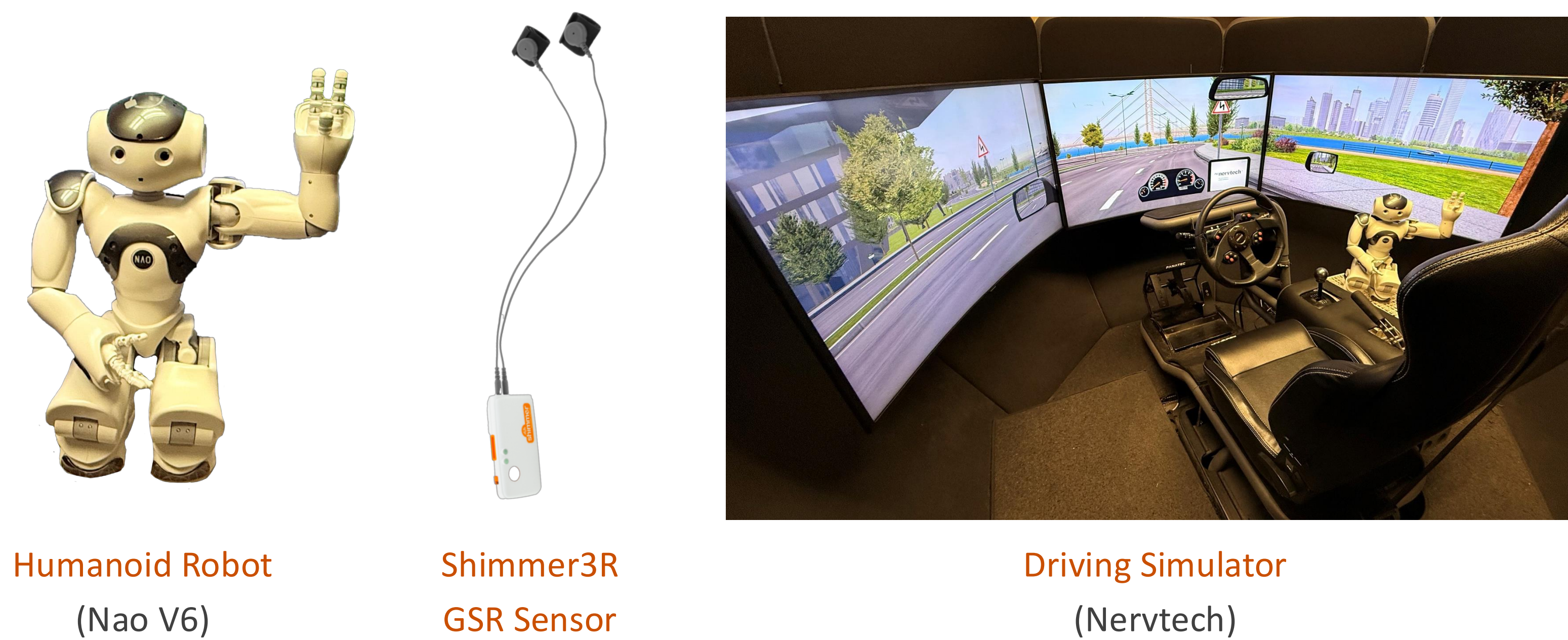
- A total of 72 participants
- 18+ years old, licensed drivers with 2+ years of driving experience

Study Design

- A mixed design study
 - One between-subject (empathy) and one within-subject (interactivity) factor
- Each driving scenario includes 6 road rage-related events.
- Participant randomly assigned to three different empathy types during driving



Apparatus

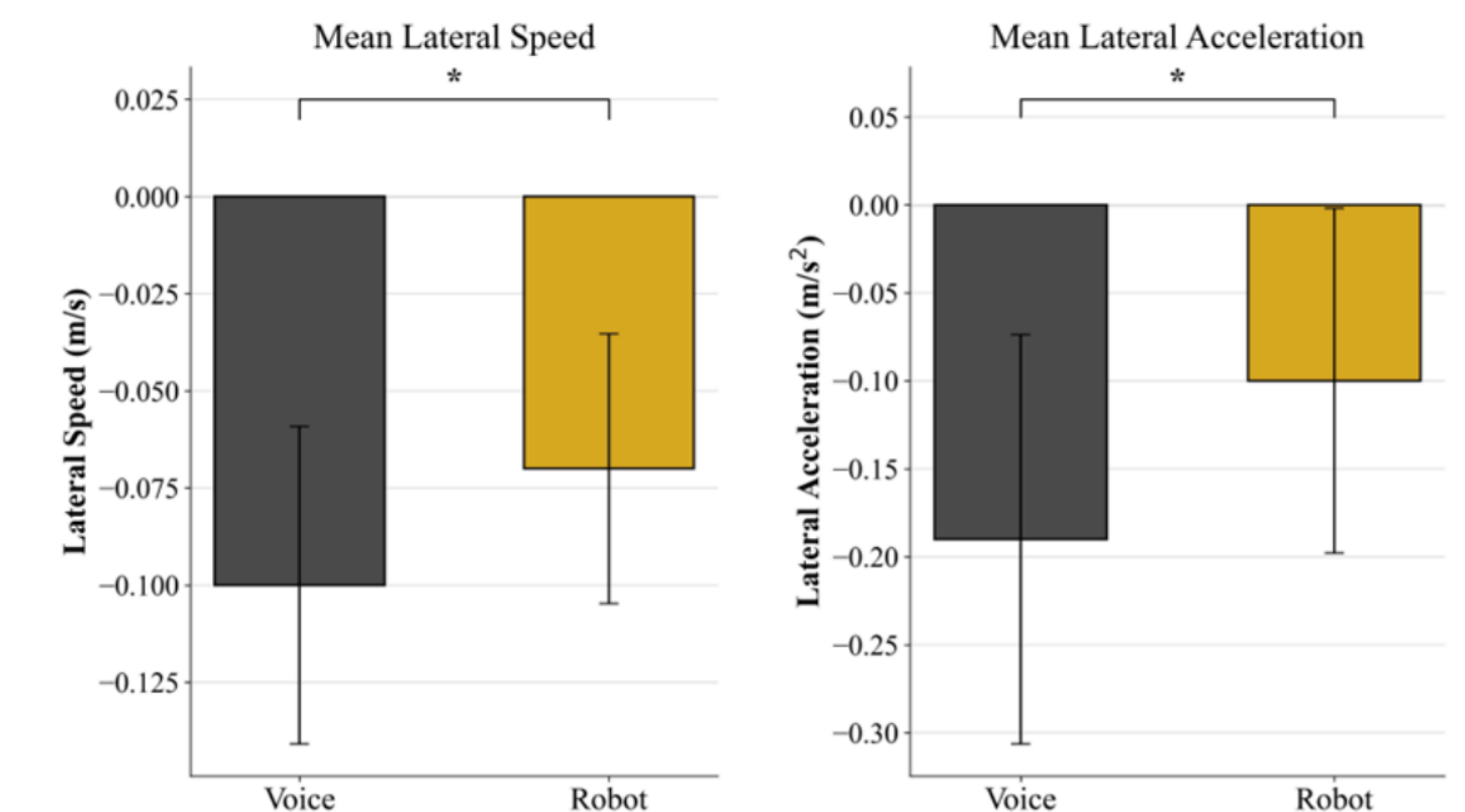


Measures

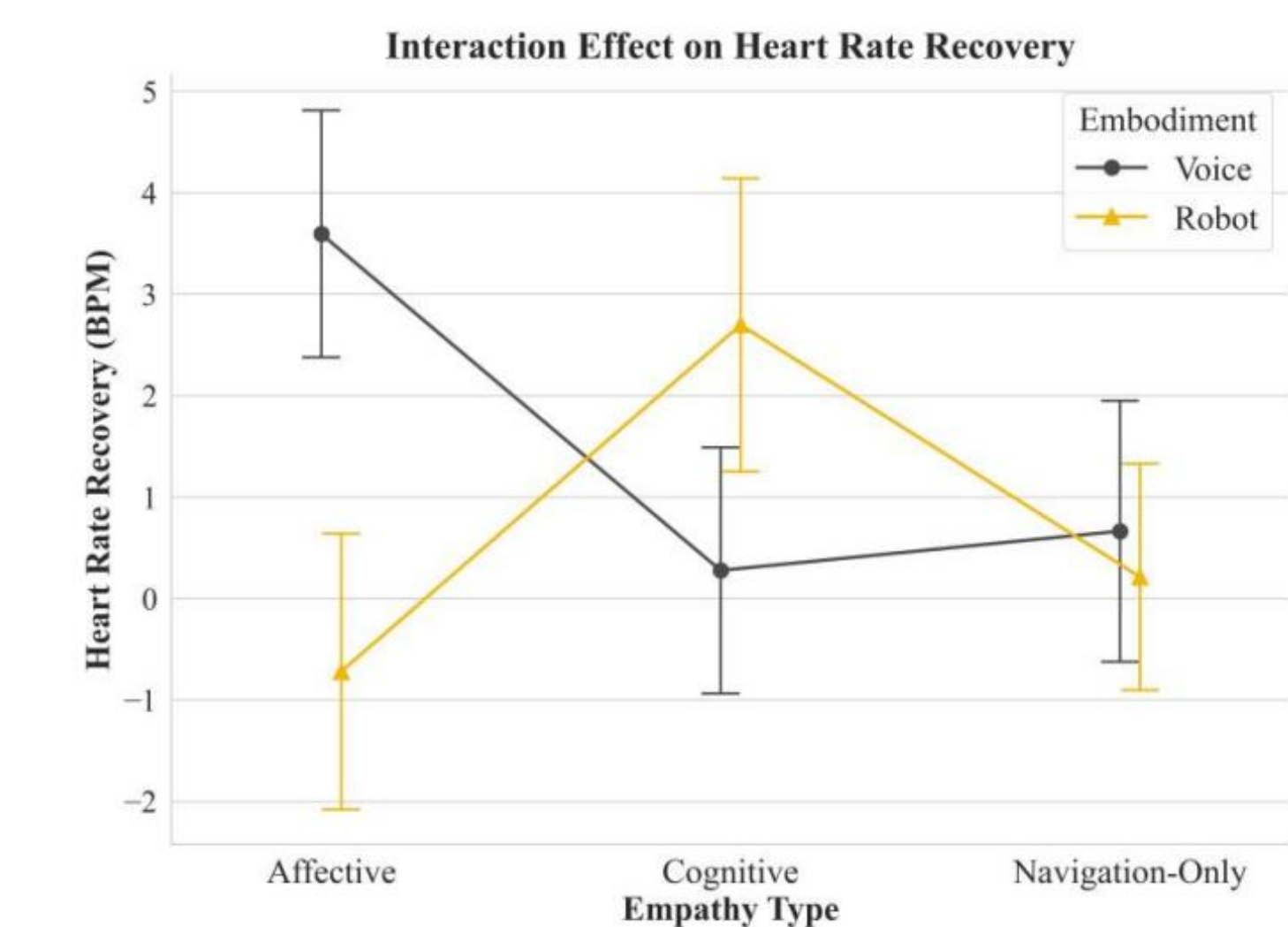
- Driving performance (e.g., speed, acceleration, brake pedal force, steering wheel torque, and lane deviation)
- Physiological reaction (e.g., HR, HRV, EDA)
- Subjective workload (NASA TLX)
- Subjective perception rating (e.g., Likeability, animacy, anthropomorphism, competence, warmth, discomfort, and trust)
- Emotional States (Positive And Negative Affect Schedule)

Preliminary Results

- Embodied robot agent led to a *more stable driving performance*



- Affective empathy was associated with *higher HR recovery* in the robot condition, suggesting it may require a physical interface to connect with the agent



Expected Results

LLM-powered agent with affective empathy:

- May yield higher ratings of warmth, naturalness, and likeability compared to pre-scripted agents
- May enhance perceived social presence and trust through more dynamic, human-like interaction

LLM-powered agent with cognitive empathy:

- May still be effective, but could be perceived as slightly less emotionally engaging compared to affective style

Effects of conversational interactivity:

- Increased interactivity may lead to greater engagement and perceived intelligence of the agent
- Conversational capability may offload cognitive demand by allowing drivers to receive clarification and reinforcement,